

Classes sharing identical feature vectors

Row: original label Column: other label on the same feature vector Cell: share of all rows in the row class (%)

CIC2018

Conflicting rows 26.65%

	Ben	Bot	Brute	DDoS	DoS	Inf	Web
Benign				.	.	26.7	.
Bot							
Brute force					67.2		
DDoS	.						
DoS	.		34.8				
Infiltration	75.1						.
Web attack	69.5					17.0	

ToN-IoT

Conflicting rows 60.73%

	Back	Ben	DDoS	DoS	Inj	MitM	Pass	Ran	Scan	XSS
Backdoor		1.3	.	.	.	.	.	.	.	.
Benign	2.3		13.7	8.6	.	8.2	.	2.4	49.9	3.3
DDoS	.	76.3		.	75.0	.	22.4	.	1.4	69.1
DoS	.	33.1	12.8		12.2	1.6	.		9.1	26.3
Injection	.	7.5	3.4	2.0		.	2.1	.	2.0	9.0
MitM	.	8.5	3.1	7.5	5.3		2.1		2.8	7.2
Password	.	33.3	.	.	.	.			.	.
Ransomware	.	9.2	.		.				1.9	1.0
Scanning	.	97.1	.	.	.	.	.	3.6		.
XSS	.	37.0	12.2	6.8	29.4	.	15.4	.	.	

BoT-IoT

Conflicting rows 6.11%

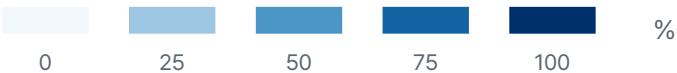
	Ben	DDoS	DoS	Recon	Theft
Benign		.	7.7	63.8	
DDoS	.		1.6	.	
DoS	1.2	2.0		1.2	
Recon	35.6	.	2.5		1.4
Theft				67.6	

UNSW-NB15

Conflicting rows 1.33%

	Anal	Back	Ben	DoS	Expl	Fuzz	Gen	Recon	Shell	Worm
Analysis										
Backdoor				16.7	18.1	18.7	19.1	17.1	13.7	.
Benign										
DoS		8.6			9.2	9.1	8.4	8.6	7.5	
Exploits		4.0		3.9		6.0	4.6	4.7	3.6	.
Fuzzers		14.0		13.4	18.0		15.2	15.8	13.0	.
Generic		66.6		.	4.6	1.4		.	.	.
Recon		29.6		29.4	30.5	31.0	3.9		3.4	
Shellcode		11.8		3.3	5.8	6.6	3.5	3.7		
Worms		4.4			.	.	4.4			

Blank: 0% .: below 1% Grey: same class A row can count in several cells, so cells do not add up



Web attack: 1,764 + 432 – 429 overlap = 1,767 rows removed · 771 remain

Class distribution after cleaning

All data (train + validation + test) · row counts and within-dataset class share (%)

CIC2018

Class	Before	After	Share before → after
Benign	17,514,626	12,798,328	87.07 → 86.74
Bot	207,703	207,703	1.03 → 1.41
Brute force	575,194	188,414	2.86 → 1.28
DDoS	1,324,350	1,316,025	6.58 → 8.92
DoS	302,966	197,415	1.51 → 1.34
Infiltration	188,152	46,761	0.94 → 0.32
Web attack	2,538	771	0.013 → 0.005
Total	20,115,529	14,755,417	100 → 100

Infiltration down 75.1%
Web attack down 69.6% · 771 rows after cleaning

ToN-IoT

Class	Before	After	Share before → after
Backdoor	203,384	200,688	0.74 → 1.86
Benign	16,792,214	6,446,120	61.02 → 59.65
DDoS	4,141,256	925,247	15.05 → 8.56
DoS	203,456	113,655	0.74 → 1.05
Injection	381,777	341,263	1.39 → 3.16
MitM	6,013	5,446	0.022 → 0.050
Password	1,594,777	1,061,699	5.79 → 9.82
Ransomware	3,971	3,604	0.014 → 0.033
Scanning	1,358,977	39,069	4.94 → 0.36
XSS	2,834,435	1,670,497	10.30 → 15.46
Total	27,520,260	10,807,288	100 → 100

Global model and residual expert structure

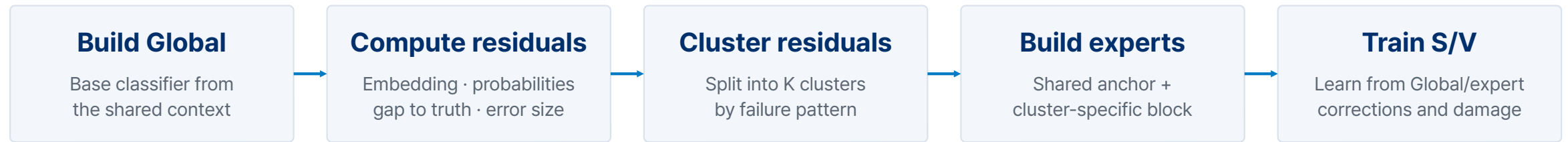
Separate roles of the training data · Global construction / expert construction / S-V training

Global pool · shared context

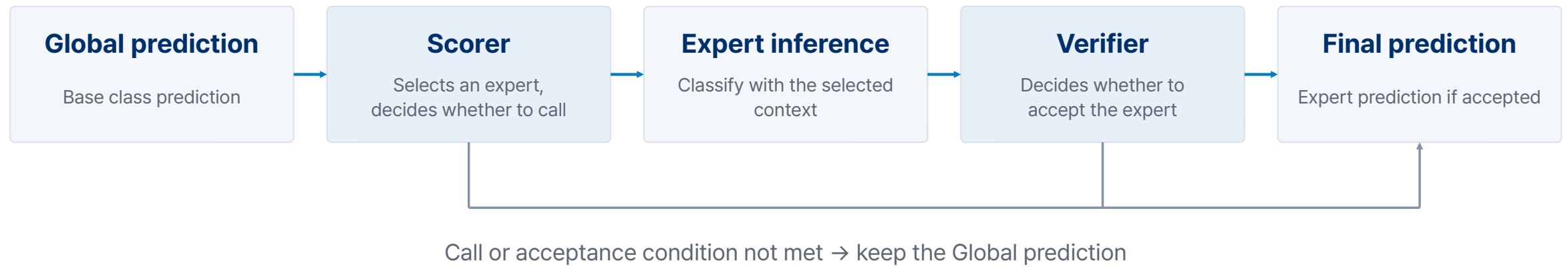
Expert pool · residuals and experts

Route pool · S/V training

Offline · model construction

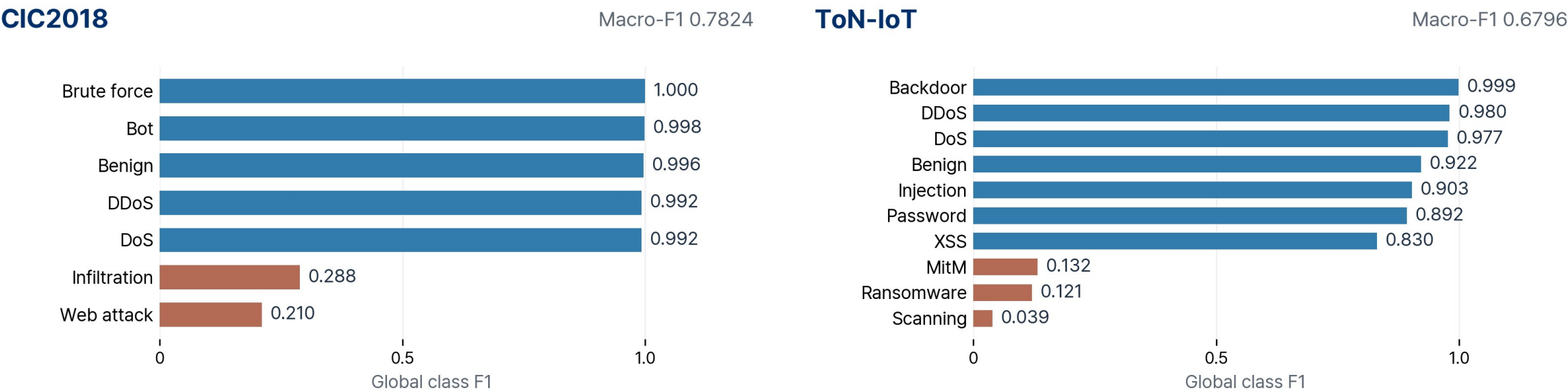


Online · per-input prediction



Global per-class performance and the expert bottleneck

Cleaned data · seed 43 · diagnosis of the existing residual contexts · before S/V



Existing e3 · Infiltration

Own residual region F1 0.440 → 0.967

Input-nearest group precision 12.4% · FP 7,935 → 21,535

Existing e3 · Injection

Own residual region F1 0.159 → 0.939

Input-nearest group precision 8.7% · FP 581 → 14,222

Own region: assigned by true residual · Input-nearest group: assigned by input embedding and Global probabilities · Arrow: Global → Expert

SOTA classification on cleaned data

Seed 43 · same C0 100,000 rows · CIC2018 test 3.04M rows / ToN test 2.27M rows

Method	CIC2018 Macro-F1	ToN Macro-F1	Training pool rows · CIC / ToN
Global TabPFN v3	0.7824	0.6796	100,000
XGBoost	0.7821	0.6983	100,000
BoostPFN	0.7148	0.5866	100,000
LoCalPFN · FT	0.7024	0.6462	100,000
DistPFN · v3	0.7839	0.6682	100,000
XGBoost · full train	0.7795	0.7162	8.67M / 4.67M

Full-train XGB: reference with extra training data · Global: base classifier before experts and S/V

CIC2018 · per-class differences · shared 100k

Best Infiltration F1 0.3055 · DistPFN
Best Web attack F1 0.3321 · XGB

ToN · per-class differences · shared 100k

Best MitM F1 0.2363 · XGB
Best Scanning F1 0.3281 · BoostPFN

SOTA training and inference cost

RTX 4090 24GB · up to 2 jobs in parallel · elapsed time including resource contention

Method	CIC fit (s)	CIC predict (s)	ToN fit (s)	ToN predict (s)	GPU GiB CIC / ToN
Global TabPFN v3	36.4	1,099.3	36.7	821.7	2.92 / 2.94
XGBoost	2.3	1.0	55.4	1.0	0.55 / 0.53
BoostPFN	98.5	2,308.5	97.8	1,737.6	0.89 / 0.89
LoCalPFN · FT	3,196.3	17,300.8	5,783.1	11,144.8	7.54 / 7.54
DistPFN · v3	36.4	1,099.4	36.7	821.9	2.92 / 2.94
XGBoost · full train	107.8	1.0	143.2	1.1	2.20 / 1.59

LoCalPFN fit includes validation · full-test inference includes kNN search

DistPFN shares the Global fit and inference · prior correction adds about 0.19 s

GPU: per-process peak sampled every 1 s · not a standalone speed comparison

Expert ability: own region vs input-nearest group

Draft · TBD

View A · own residual region

Representative expert · class: TBD

Metric	Global	Expert
Precision	TBD	TBD
Recall	TBD	TBD
F1	TBD	TBD

Corrections: TBD

Damage: TBD

View B · input-nearest group

Representative expert · class: TBD

Metric	Global	Expert
Precision	TBD	TBD
Recall	TBD	TBD
F1	TBD	TBD

Corrections: TBD

Damage: TBD

Takeaway TBD

Contribution of scorer and verifier

Draft · TBD

Configuration	CIC2018	ToN-IoT	Calls · accepted	Corrections · damage
Global	TBD	TBD	TBD	TBD
No S·V	TBD	TBD	TBD	TBD
S only	TBD	TBD	TBD	TBD
V only	TBD	TBD	TBD	TBD
S+V	TBD	TBD	TBD	TBD
Context · gate settings	TBD			
Takeaway	TBD			

Final performance and cost of the proposed model

Final comparison of the validated expert and S/V configuration

Configuration	CIC2018	ToN	Training info	Call cost
Expert · no S/V	TBD	TBD	TBD	TBD
Expert · S only	TBD	TBD	TBD	TBD
Expert · V only	TBD	TBD	TBD	TBD
Expert · S+V	TBD	TBD	TBD	TBD

Baselines: SOTA performance (slide 5) · measured cost (slide 6)

Includes the extra training rows for experts and routing, and the conditional inference cost

Confirmed effects and next improvements

Draft · TBD

Context composition	Expert discrimination	Final model performance
TBD	TBD	TBD
Next improvements		
TBD		
Conclusion	TBD	